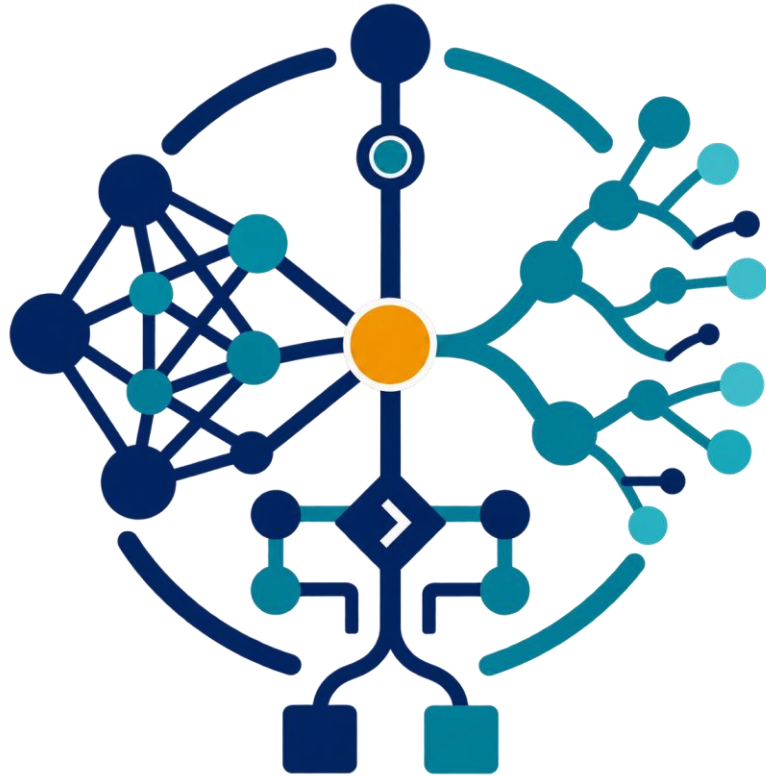




Artificial Intelligence Fundamentals

INTRODUCTION TO THE COURSE
(SOME THOUGHTS + USEFUL INFO)

About me

- andrea.cossu@unipi.it
- <https://www.andreacossu.com/contact>

Assistant professor / researcher @ CS department

- continual lifelong learning
 - recurrent neural networks
- evolutionary optimization

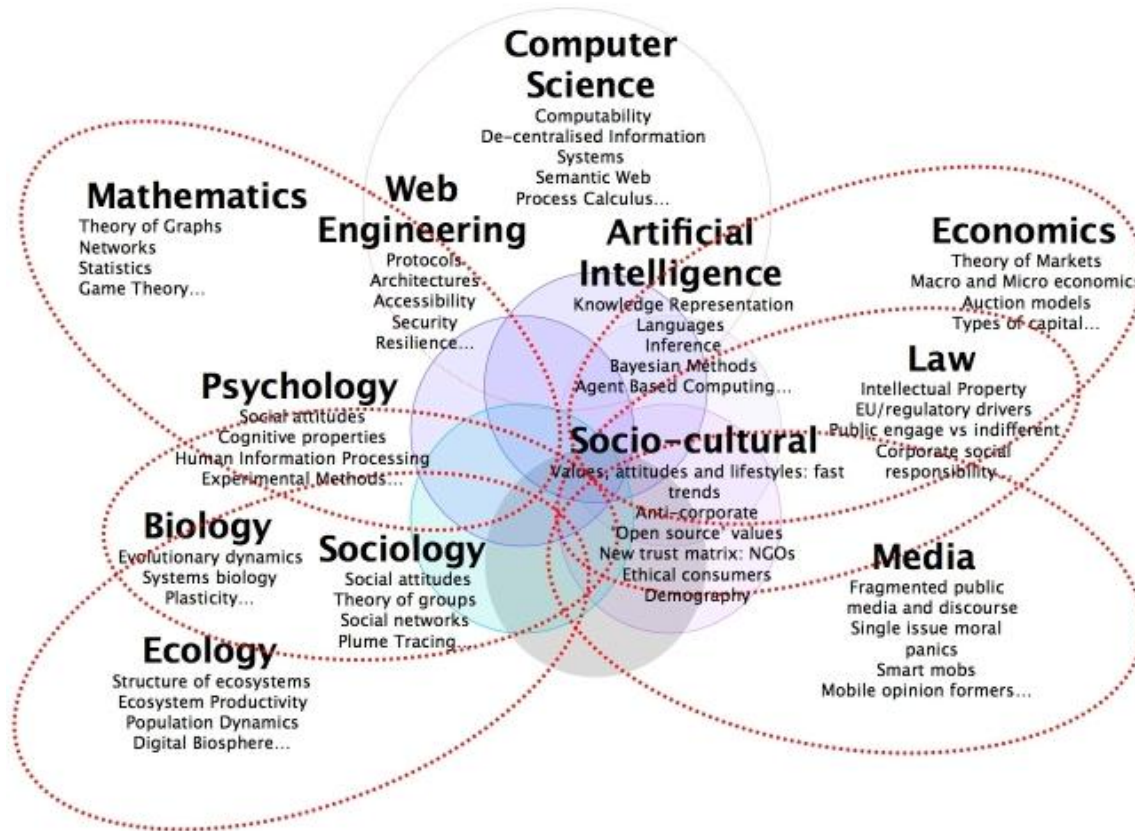
Drop me a mail for an appointment



The AI science

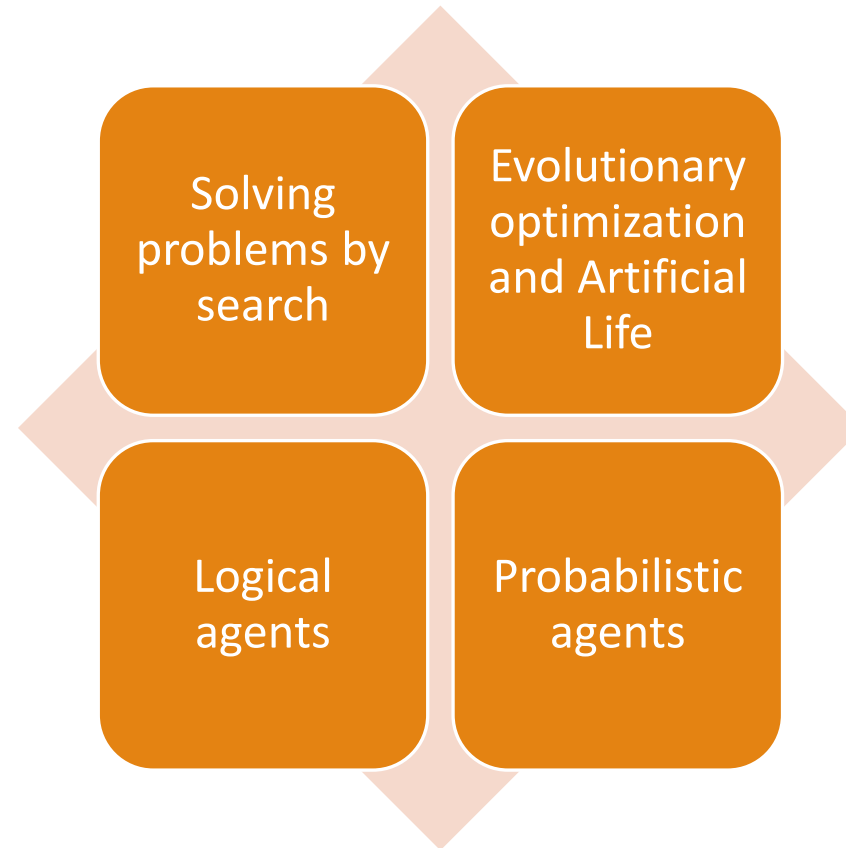
A truly multidisciplinary effort

- Math, computer science, engineering, physics: formalize and implement abstract processes
- Psychology: models of cognition, multi-agent interaction, decision-making, ...
- Neuroscience: well...
- Biology: robots inspired by living organisms, homeostasis, ...
- (Computational) Linguistics: the structure of language and how to exploit it
- Philosophy: consciousness and the sense of I, real vs. simulated
- More?



The Web Science «Butterfly»

Course topics



Course plan

1. This intro 😊
2. AI agents architecture + project
3. Uninformed and Informed search
4. Local search
5. Constraint Satisfaction Problem
6. Searching in games
7. Lab 1
8. Evolutionary Optimization
9. Novelty-based search and Quality-Diversity
10. Artificial Life
11. Lab 2
12. Inference in propositional logic
13. Inference in First-Order Logic
14. Planning algorithms
15. Bayesian Networks
16. Hidden Markov Models
17. Multi-Agent systems
18. Wrap-up + project presentations/poster

More in detail...

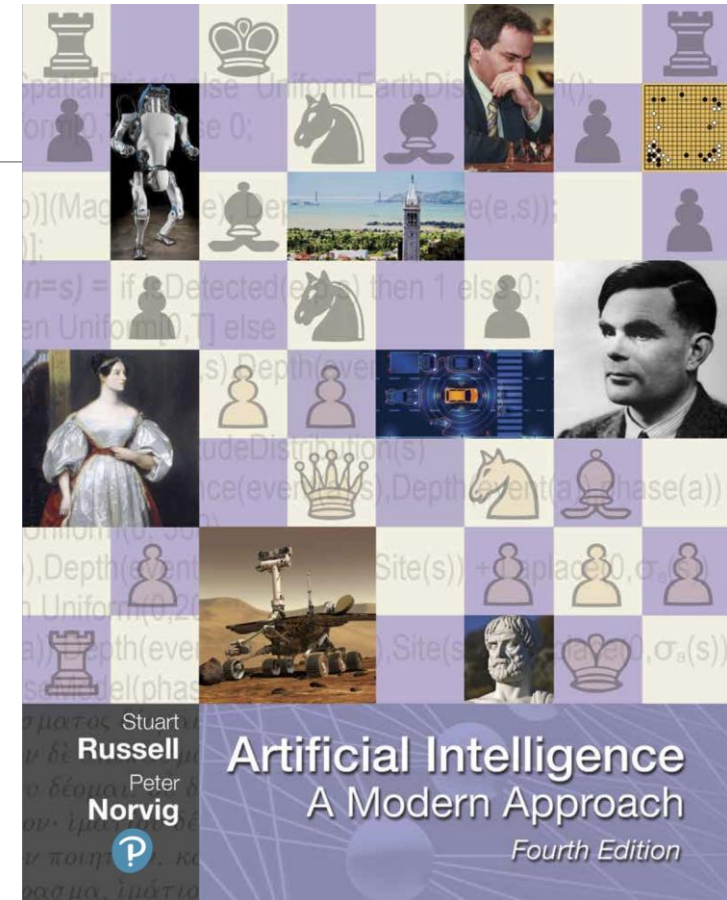
Part I, II, III, IV of AIMA book.

The book is optional, the material on Moodle is sufficient

- Search (part II)
- Knowledge representation (part III)
- Planning (part III)
- Uncertainty and probabilistic reasoning (part IV)

Moodle URL

<https://elearning.di.unipi.it/course/view.php?id=1158>



<https://aima.cs.berkeley.edu/>

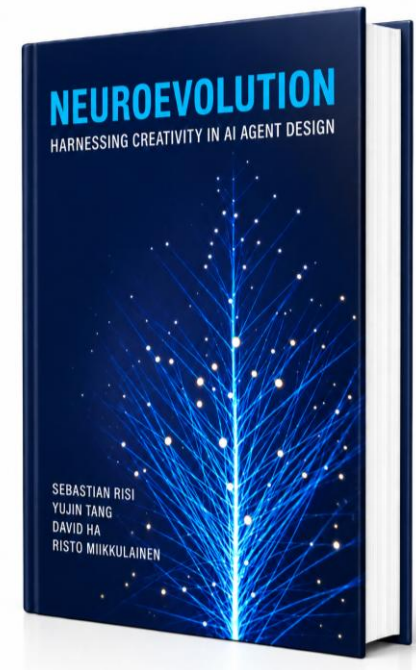
More in detail...

Evolutionary computation and Artificial Life

- The neuroevolution book
- Slides on Moodle are sufficient
- Specific chapters/sections linked in the slides

Moodle URL

<https://elearning.di.unipi.it/course/view.php?id=1112>



<https://neuroevolutionbook.com/>

First year

Semester 1	CFU	Semester 2	CFU
Artificial intelligence fundamentals	6	Generative and deep learning	9
Computational mathematics for learning and data analysis	9	Parallel and distributed systems: paradigms and models	9
Machine learning	9	Group: AI elective 9 CFU	9
		Group: AI elective 6 CFU	6
	24		33

Second year

Semester 3	CFU	Semester 4	CFU
<i>Human language technologies (not offered in the a.y. 25/26)</i>	9	Group: AI elective 6 CFU	6
<i>Computer vision (not offered in the a.y. 25/26)</i>	9		
Group: AI elective 6 CFU	6		
Group: free choice	9	Thesis	24
	33		30

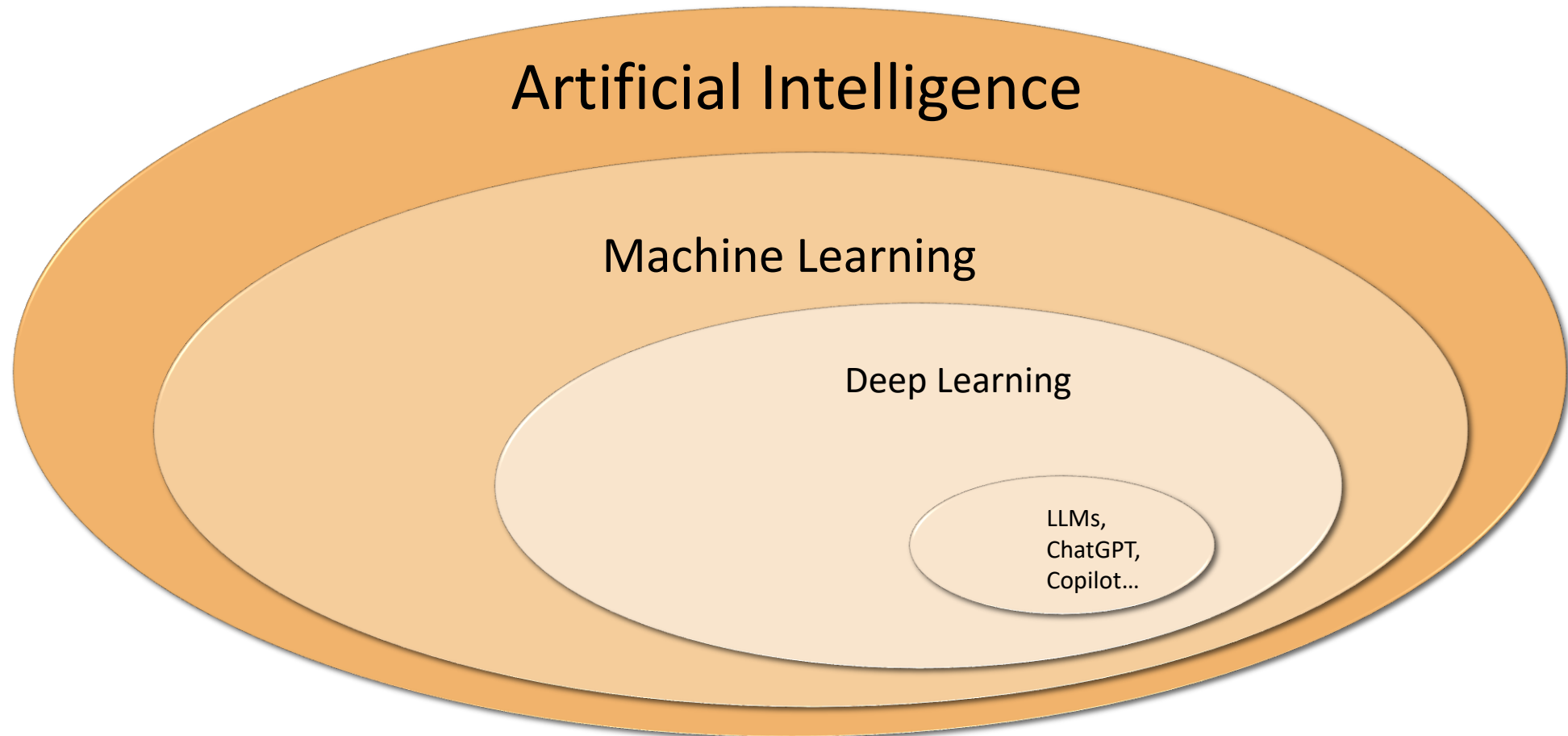
Group: AI electives (9 CFU)

Algorithm engineering (Sem. 1)
Algorithm design (Sem. 2) (*)
Data mining (Sem. 1)
Digital Health lab (Sem. 2) (*)
Mobile and cyber-physical systems (Sem. 2)

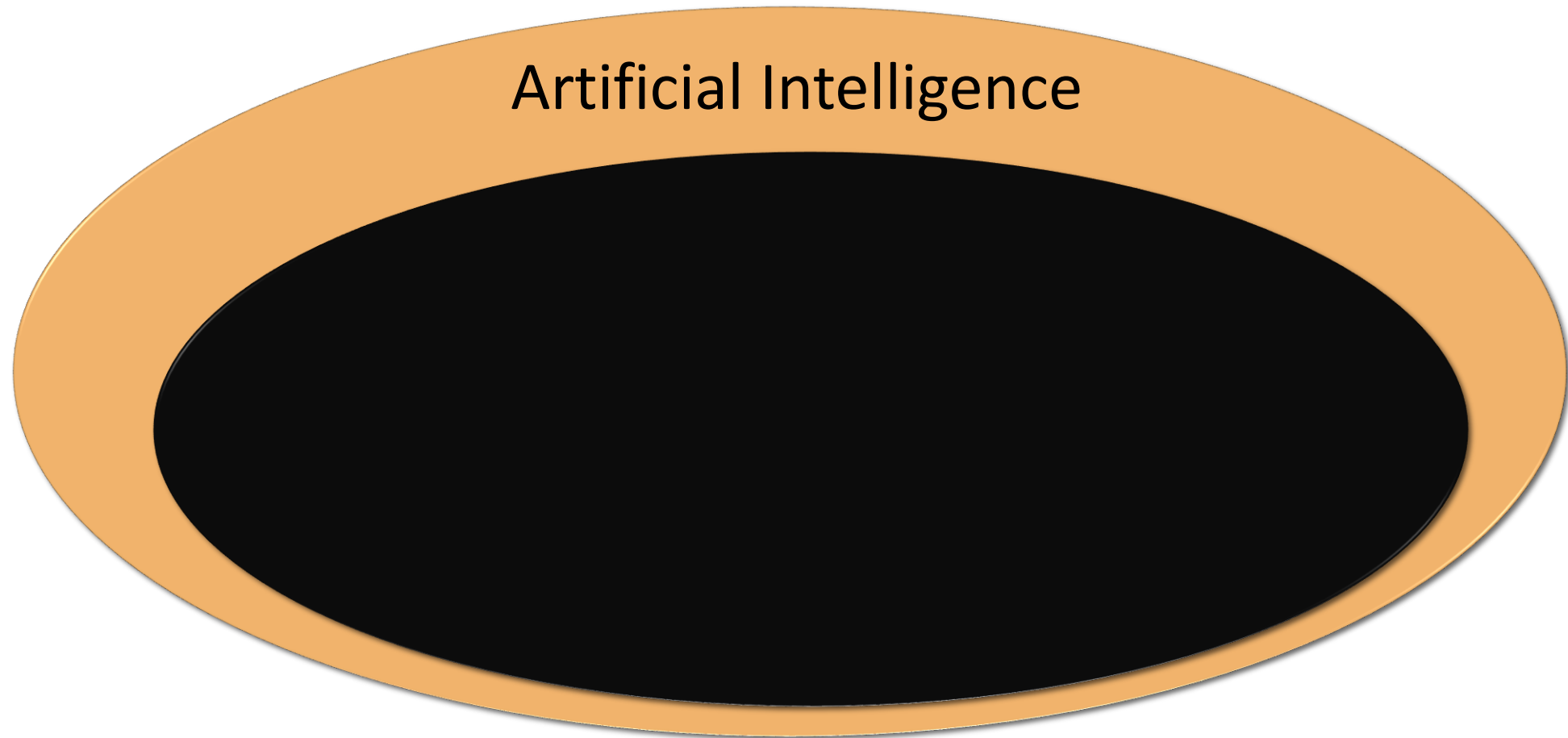
Group: AI electives (6 CFU)

3D Geometric Modeling & Processing (Sem. 1)
Accelerated Computing (Sem. 1)
Algorithmic Game Theory (Sem. 2)
Competitive programming and contest (Sem. 1) (*)
Computational models for complex systems (Sem. 2)
Computational neuroscience (Sem. 2)
Continual learning (Sem. 2)
Information retrieval (Sem. 1)
Introduction to Quantum Computing (Sem. 2)
Laboratory on ICT Startup Building (Sem. 1)
Learning on Graphs (Sem. 1) (*not offered in the a.y. 25/26*) (*)
Robotics (Sem. 2)
Scalable Distributed Computing (Sem. 1) (*)
Scientific and large data visualization (Sem. 2)
Semantic web (Sem. 1)
Smart applications (Sem. 1) (*not offered in a.y. 25/26 – Smart Applications, 9 cfu, is offered instead*)
Social and ethical issues in computer technology (Sem. 2)

Artificial Intelligence, a large family



Artificial Intelligence, a large family





freedom from deep learning

«all» AI, *but* learning*

why? don't we love the cool stuff?

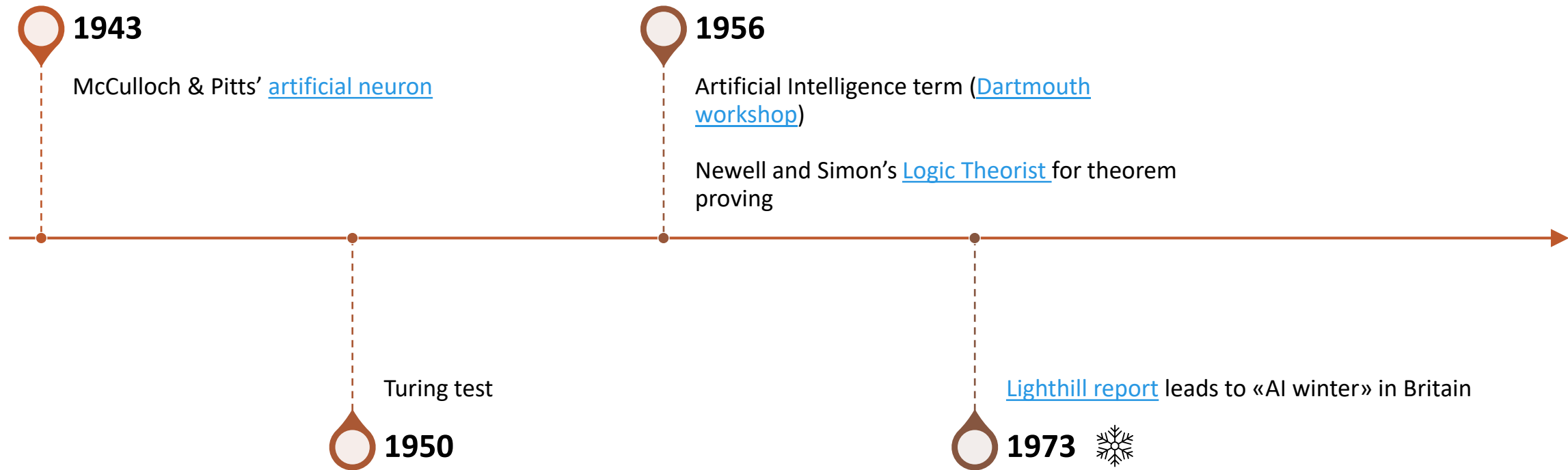
YES

* some learning might be surreptitiously introduced by the teacher



who are
you?

(some of) AI achievements and milestones



DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

A more recent controversy

 **Jürgen Schmidhuber** 
@SchmidhuberAI

 [Mostra traduzione](#)

In 1956, Cybernetics was renamed "AI" by logic programming experts. Now Trump wants to rename AI again. Let's revive the old name! Modern AI is about neural nets & deep learning, closer to Cybernetics than to 1956 AI. Plus, Cybernetics sounds better than Artificial Intelligence.

3:32 PM · 20 set 2026 · **71.537** visualizzazioni

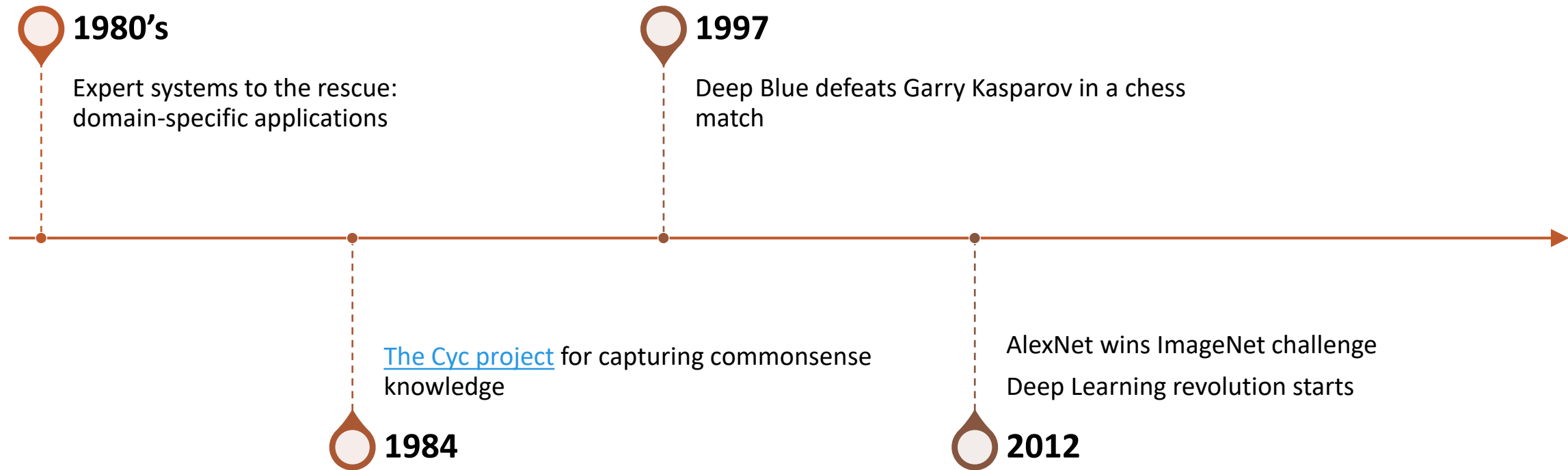
 **Donald J. Trump** 
@realDonaldTrump

 [Mostra traduzione](#)

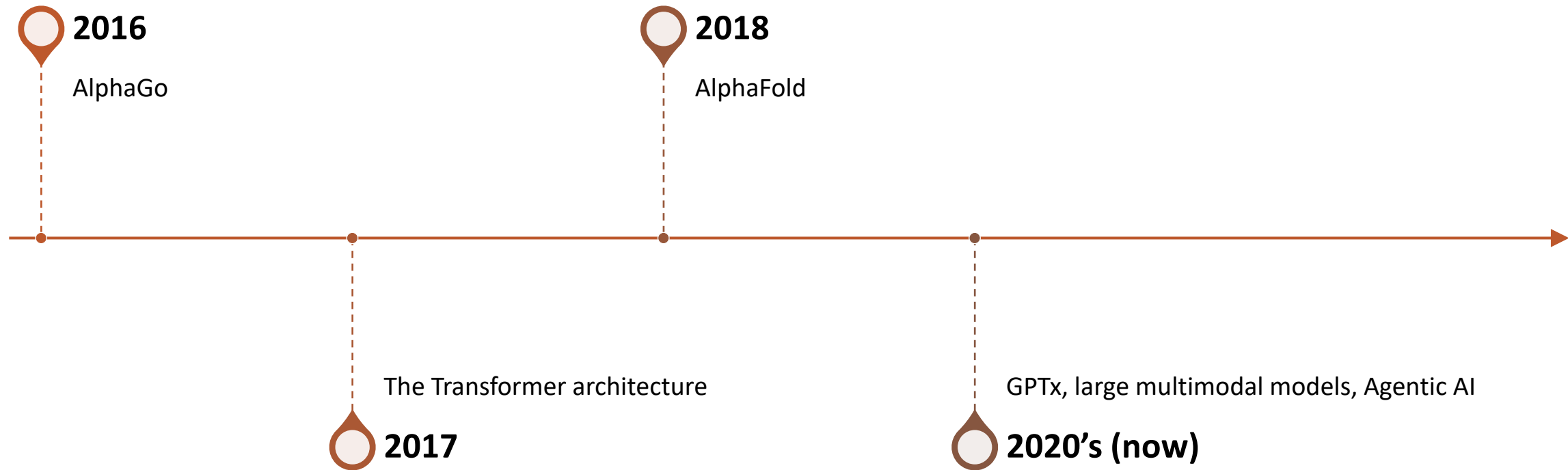
Many people think that the words "Artificial Intelligence" are inaccurate, and very ineloquent, relative to AI, or Artificial Intelligence. A far more elegant and accurate description of this new phenomena would be Superior Intelligence (SI) or, Extreme Intelligence (EI) or, Supreme Intelligence (SI). This is a Poll, and I would appreciate everybody voting! Which is the best name for this ever growing "Revolution?" President DONALD J. TRUMP

6:39 PM · 19 set 2026 · **13,5 Mln** visualizzazioni

(some of) AI achievements and milestones



Recent timeline



Symbolic AI

The dominant AI paradigm from the 50s to the 90s.

- High-level, *human-readable, interpretable* representation of the world

Symbol is a meaningful pattern that can be manipulated

Where do symbols come from?

- Humans provide symbols to computers (chicken-egg problem)
- Computers build low-level symbolic representations (sub-symbolic manipulation)
- It reminds me of Hofstadter's strange loops (books recommendations at the end)

[Semiotics](#) is the study of signs, symbols, and signification as communicative behavior

- relationship between signifier vs. signified

Symbol grounding problem

[Harnad, 1990] What/who/how to connect symbols and the corresponding real-world objects?

A symbol represent something outside the system

- «jumping out of the system» (Hofstadter)

Problematic for formal logic (Gödel's incompleteness theory)

- You have to jump out of the system to prove at least one statement of any sufficiently expressive formal system

PSSH: physical symbol system hypothesis

*“A physical symbol system has the **necessary and sufficient** means for intelligent action”*
[Allen Newell, Herbert A. Simon]

Humans are physical symbol systems, so are digital computers as both manipulate symbols.

Physical because symbols are manipulated to produce other symbols

- it is a physical process that consumes energy

Is AI conscious then?

Not so fast, intelligence $\neq$ consciousness

- Consciousness is often linked with subjective experience

How to get that? How to *know* our agent has a subjective experience?

- “What Is It Like to Be a Bat?” Thomas Nagel
- Dissecting the system and studying its behaviour may not be enough

How to distinguish between me and a **zombie**? (Searle’s [Chines room](#))

On the other hand, *if* there is consciousness, it must be a byproduct of the “body”.
It must be “somewhere” (materialism)

- Dualism considers mind a different substance than body

Does consciousness even require a body?

- Hofstadter contends it’s the *symbolic pattern* that is paramount, not the nature of the physical substrate. That is, if their logical activity was organized in an equivalent way, silicon chips could support consciousness just as neurons do.

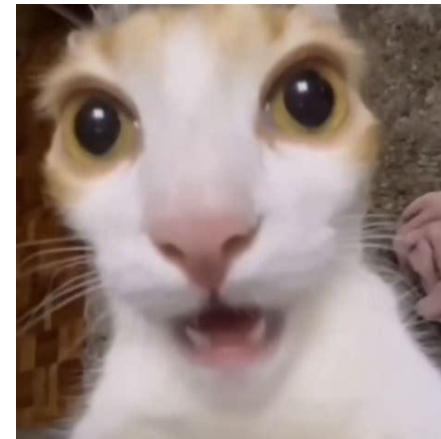
[I Am A Strange Loop by Douglas Hofstadter | Issue 78 | Philosophy Now](#)

Not so easy, is it? Let’s not dig further...

Symbolic AI: recognising a cat

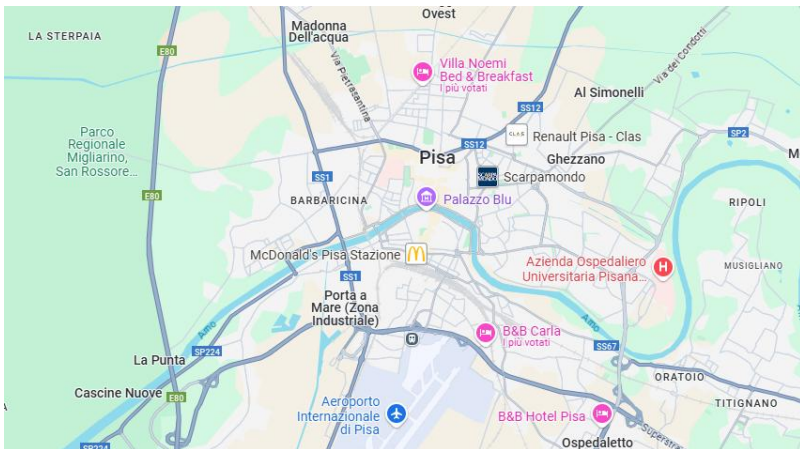


The real world is complex: sub-symbolic representations

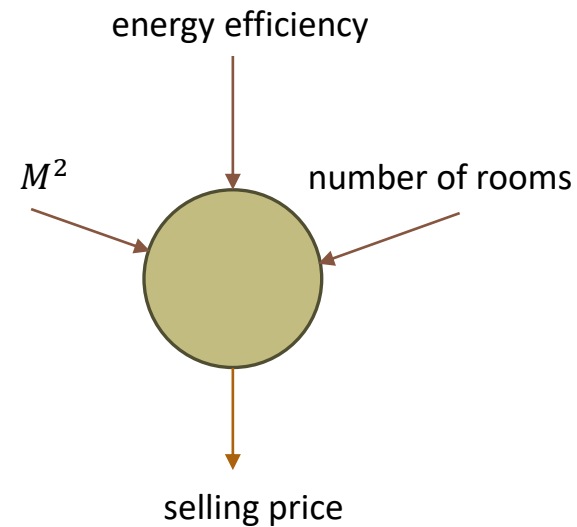


Models of the world

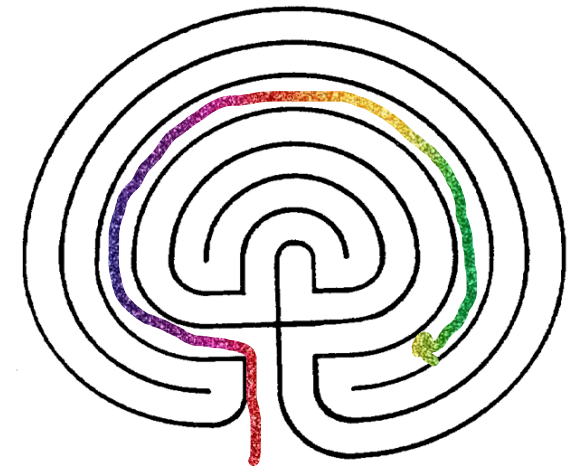
Descriptive



Predictive



Decisional



Machine/Deep learning in a nutshell

To build it (training):

Machine learning maps X to Y

- It requires **huge** amount of data
- The more complex the mapping (the task), the more data is required

To use it (inference):

Expensive computational resources (memory, time)

It's getting better, but large models are still large. And growing.

Ad-hoc accelerators (GPUs, TPUs, ...) are needed.

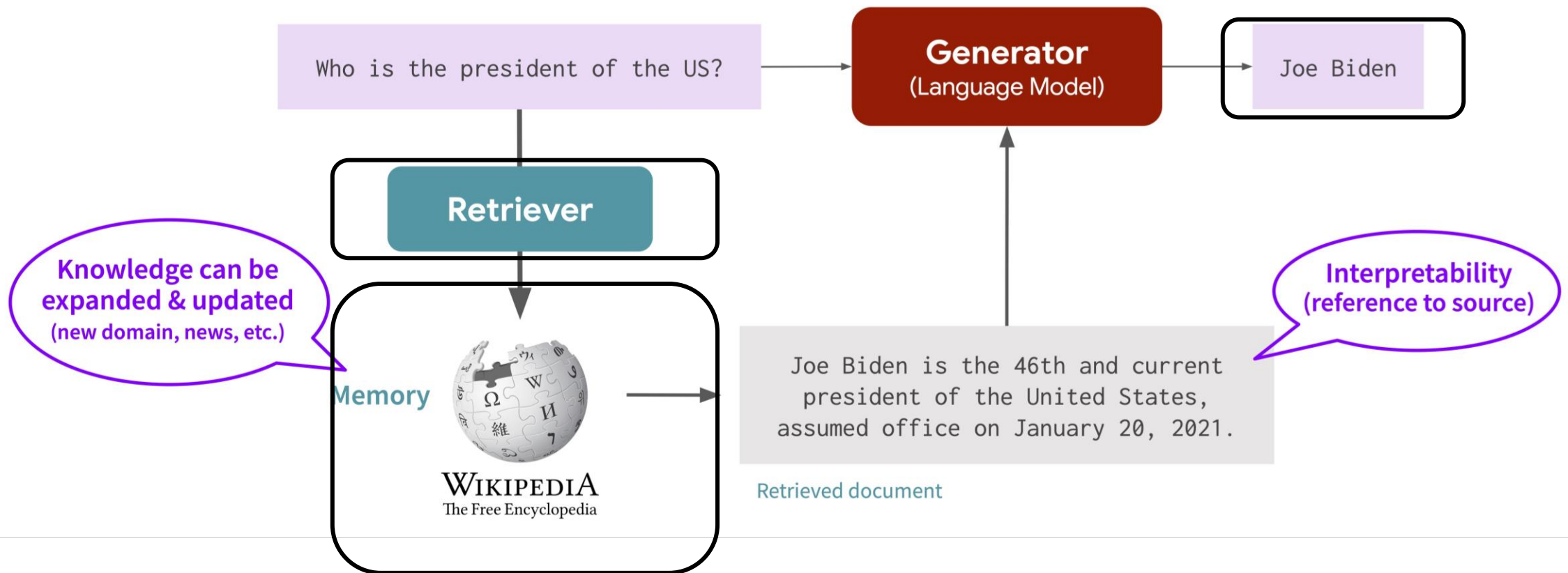
Deep learning requires even more data and compute

Hybrid systems

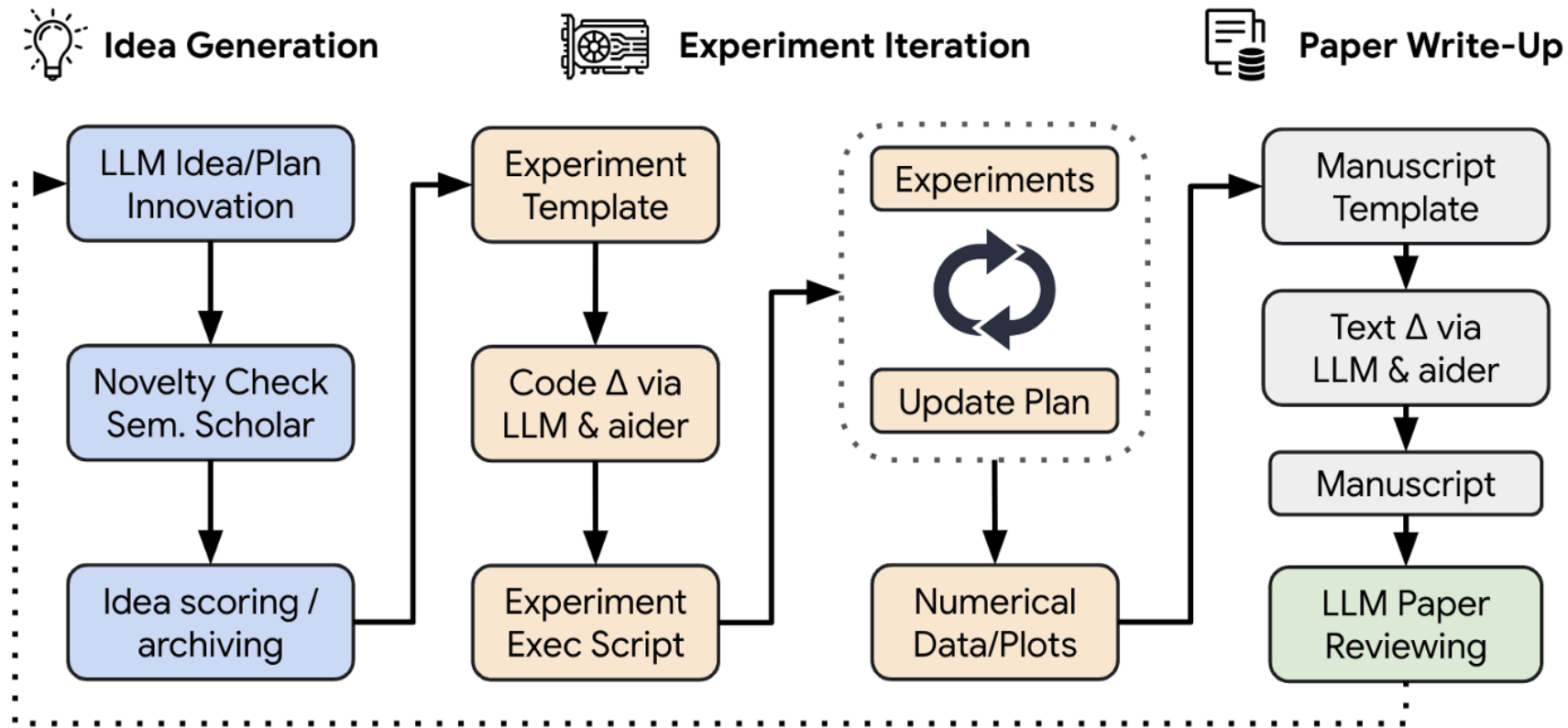
Deep learning works great in several applications

This course is not saying «don't use deep learning».
It's saying «don't use deep learning **only**»

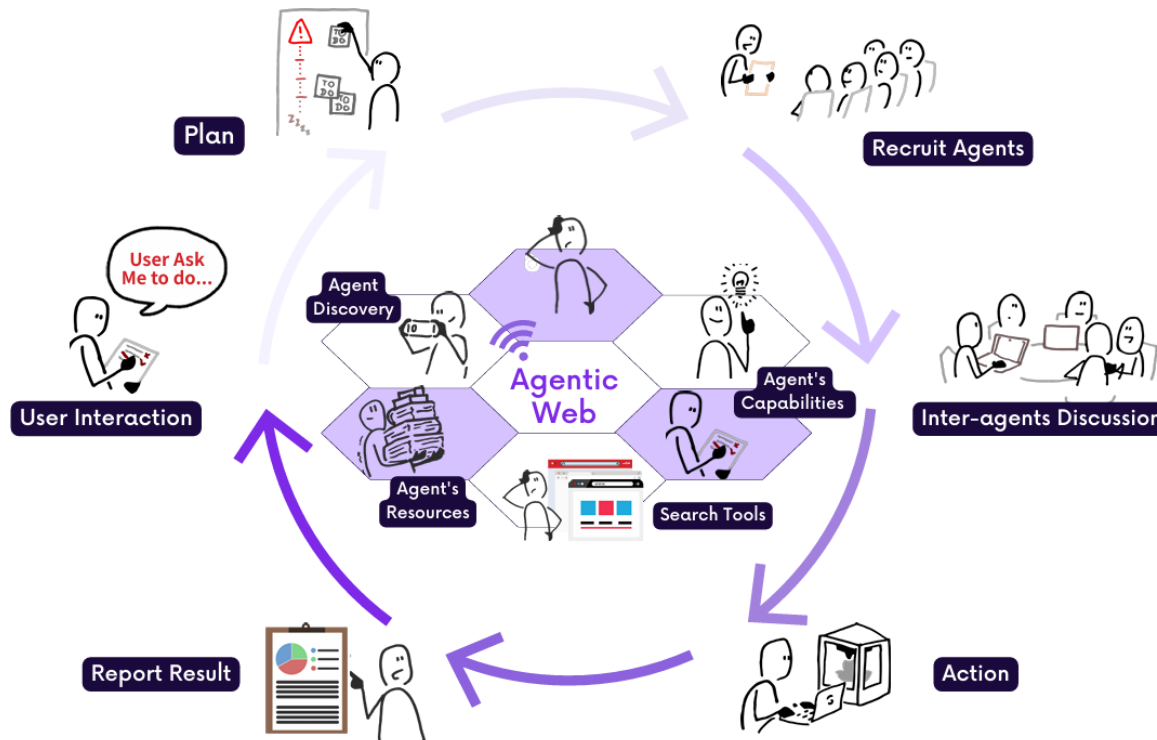
One motivating example: RAG



One motivating example: the AI scientist



The Agentic Web



Autonomous collection of information → machine-ready vs. human-ready

Machine-to-machine interaction, mostly

Who feeds the web with data?

- are we humans just data provider for intelligent agents?

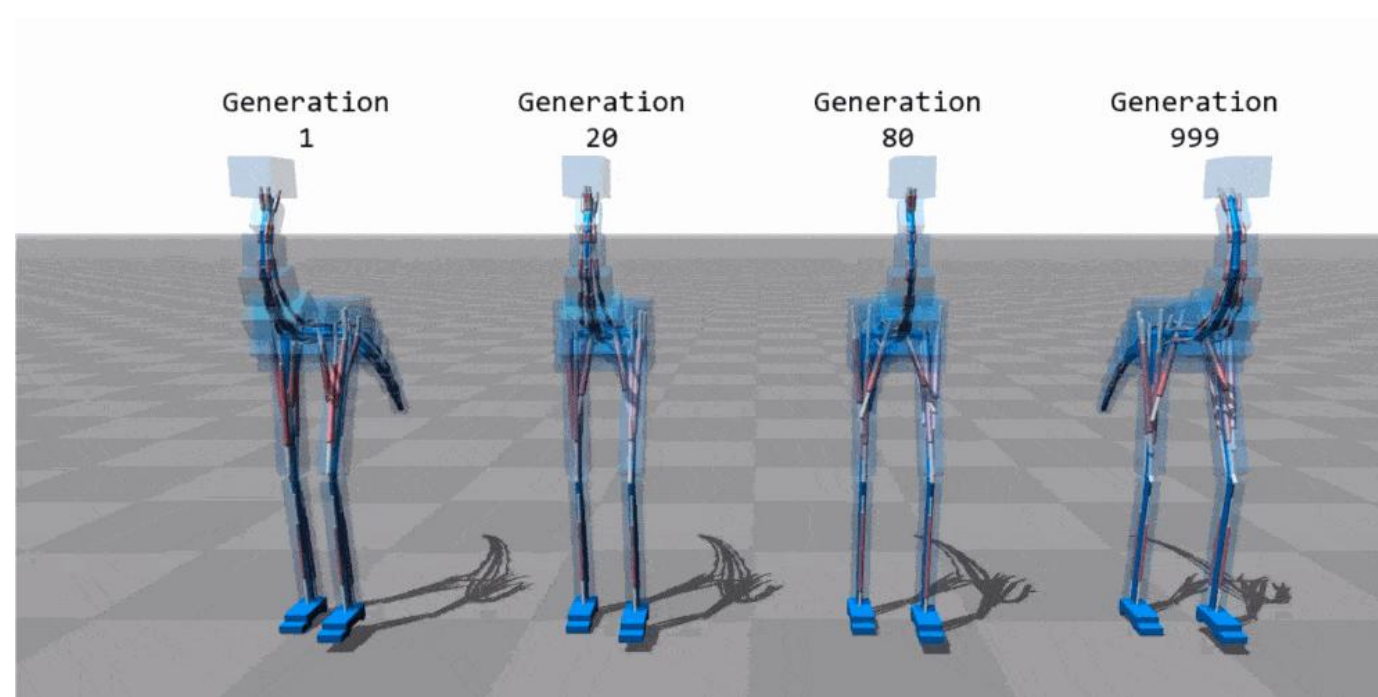
Evolutionary Computation

One of the algorithm of life

- Learning is probably another one

Can optimize for wild objectives

- What about novelty? Interestingness?



Course prerequisites

- Background from computer science
 - ✓ Algorithms and complexity
 - ✓ Basics of formal logic
 - ✓ Elements of probability and calculus

When in doubt, raise your hand!

Grading: two different ones

- **Modality 1: oral exam**

- Final Grade = oral exam on the course topics

- **Modality 2 (only attending students): oral exam + project**

- Let X be the grade of the oral exam
 - If $X \geq 18$; then final grade = 30% project + 70% X ; else fail

In case of failure you can keep the same project

- **Non-attending students → Modality 1**

Project??

It's optional. It's for you to try out things. Based on **your** initiative. **You** want to do it.

- I can/will suggest project topics

A few lectures dedicated to project development

Depending on the number of projects:

- Short project presentation towards the end of the course
- Poster presentation

Small group projects (max 3 people).

Larger groups: possible if the project is *really* challenging.

In any case, there should always be a «who does what» beforehand

Project plan + project execution + final presentation

- The execution might deviate from the plan, just explain why

Non-fictional books – for fun

Anything from Douglas Hofstadter:

- Anelli nell'io – I am a strange loop
- **Concetti fluidi e analogie creative – Fluid concepts and creative analogies (!!)**
- Gödel, Escher, Bach

Daniel Dennett: L'idea pericolosa di Darwin – Darwin's dangerous idea

Richard Dawkins: Il gene egoista – The selfish gene

Antonio Damasio: L'errore di Cartesio – Descartes' error

David Chalmers: Più realtà - Reality+: Virtual Worlds and the Problems of Philosophy

...

I am accepting suggestions as well

Make-up lectures

To be scheduled when needed. Announcements will be published in Moodle course page

What about Monday 14-16?

- Pending room availability



Upcoming – Intelligent agents